

WESTFIELD REFERENCE RESEARCH · DRAFT REVIEW

Expandable Concertina Cooler

A compact reference deck for foldable, collapsible, and expandable cooler architectures. The goal is to identify practical mechanisms, manufacturable material systems, and risks before concept development.

Design Direction

The working product idea is a cooler that can collapse for storage and expand when the user needs more volume, using a concertina or bellows-like structure. The strongest path appears to be a hybrid construction: controlled folding side structure, segmented insulation, leak-resistant liner strategy, and a carry load path that does not rely on the folding wall.

What the market already validates

Consumers understand the value of compact storage, especially for cars, apartments, RVs, beach gear, grocery runs, sports sidelines, and tailgating. The key opportunity is making collapsibility feel more premium and controlled.

What the concept must solve

A true concertina cooler has to manage insulation gaps, water collection in fold valleys, liner slack, zipper alignment, and handle loads. These are manufacturing problems as much as styling problems.

Reference Set

These examples are not all direct competitors. Some are mechanism references, some are manufacturing references, and some are useful because they show what commodity fold-flat coolers already do.



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COLLAPSED

ORIGAMI SOFT FORM

BUILT Origami Neoprene Tote

Photo reference supplied for the BUILT Origami Neoprene Wine Bottle Tote.

Small neoprene bottle tote that folds from a flat soft form into volume through origami-like geometry.

- Good reference for fold language and compressibility.
- Limited structural and payload relevance for a larger cooler.
- Shows how soft material stretch can hide pattern complexity.



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SEMI-RIGID PANELS

Kelty Folding Cooler

Foldable cooler using fabric, molded PE foam panels, reinforced bottom details, and removable PEVA liner.

- Strong reference for cooler-like structure in a folding format.
- Panel strategy is more manufacturable than continuous insulated bellows.
- Removable liner simplifies cleaning and leak management.



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FOLDED STORAGE

FLEXIBLE INSULATION

Nutshell Coolers 20 L CocoFiber Cooler

Boxier 20.5 L soft cooler using coconut husk fiber insulation, recycled polyester shell materials, and fold-flat storage.

- Useful reference for a larger rectangular cooler volume with structured soft-panel walls.
- Folds to roughly half size for storage while retaining a wide flip-top opening.
- Good material-story benchmark: coconut fiber insulation, removable welded liner, and recycled outer shell.



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CONTROLLED COLLAPSE

CleverMade Tahoe

Collapsible soft cooler with SnapHinge-style side structure, metal rim, baseboard, and insulated liner.

- Clean benchmark for controlled folding and fast setup.
- Collapsed height around 2.5 in is a strong storage story.
- Rim and baseboard help the soft cooler feel intentional.



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HARD COOLER PROOF

Coleman Snap 'N Go

New collapsible hard-sided cooler family announced for 35, 45, and 55 qt formats.

- Market proof that collapsible coolers are moving into mainstream hard-cooler territory.
- Claims collapse to one-third size and quick open/collapse action.
- Useful benchmark for consumer expectations around performance.



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MASS-MARKET SOFT COOLER

Coleman 54-Can Collapsible

Large fold-flat soft cooler with leak-proof antimicrobial liner, shoulder strap, zipper, and pocketing.

- Useful commodity price and feature benchmark.
- Simple fold-flat format, not much volume adjustability.
- Good reminder that low-cost expectations are already established.

Mechanism Matrix

The references point to several viable families. Concertina should borrow from more than one rather than copying a single product.

Family
Soft origami fold
BUILT Origami Tote
Useful visual and pattern language, but weak structural precedent.
Structured folding frame
CleverMade Tahoe
Strongest soft-cooler collapse mechanism reference.
Semi-rigid panel system
Kelty Folding Cooler
Most realistic route for insulation and shape control.
Collapsible hard cooler
Coleman Snap 'N Go
Market proof and performance benchmark, though likely different tooling.
Flexible insulation
Nutshell
Potential solution for repeated fold zones and a stronger sustainability story.

Best Ideas To Borrow

Controlled fold structure

Borrow the idea of snap hinges, rims, baseboards, and defined fold paths so collapse feels deliberate instead of floppy.

Segmented insulation

Use foam or flexible insulation in bays rather than expecting a thick continuous insulated wall to concertina cleanly.

Independent load path

Route handles, shoulder straps, and trolley loads through reinforced straps, rails, or side columns instead of through the bellows wall.

Volume lock positions

Use straps, clips, zippers, snaps, or toggles to hold low, medium, and full-volume states without the cooler ballooning.

Liner strategy

Consider a removable welded bucket liner or slack-managed inner liner to keep the leak zone simpler than the outer folding body.

Base stays rigid

Keep the bottom more rigid and durable than the walls so the cooler can be loaded, dragged, or set down without collapsing unpredictably.

Manufacturing Risks

Concertina-specific issues

- Fold valleys can collect water, dirt, and ice melt.
- Repeated folding can crush foam and create thermal bridges.
- Sharp pleats are bad for welded liners and coated materials.
- Expanded height changes can disturb lid and zipper alignment.

Construction issues

- Welded seam paths need access, pressure, generous radii, and compatible material stacks.
- Too many layers at handle or zipper zones can cause cold welds, burn-through, or cosmetic defects.
- External pockets and heavy hardware add leakage and stress complications.
- Any waterproof claim needs zipper, liner, seam, and QA support.

Recommended concept path

Develop a hybrid concertina cooler: rigid or semi-rigid base, controlled sidewall fold structure, segmented insulation panels, welded removable or slack-managed liner, compression/locking straps for volume settings, and carry loads routed around the folding wall.

Source Links

- [BUILT Origami Neoprene Bottle Tote](#)
- [Kelty Folding Cooler](#)
- [Nutshell Coolers 20 L CocoFiber Cooler](#)
- [CleverMade Tahoe Collapsible Cooler](#)
- [Coleman Snap 'N Go Collapsible Hard Cooler](#)
- [Coleman 54-Can Collapsible Soft-Sided Cooler](#)